Indoor location from Wi-Fi signals

A reproducible classical-ML baseline on UJIIndoorLoc

99.91%
RF building accuracy

91.99%
XGBoost floor accuracy

6.86 m
RF median coordinate error

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One Wi-Fi fingerprint, three prediction tasks

Input

520 AP signal readings

Sparse and device-dependent

+100 means not heard

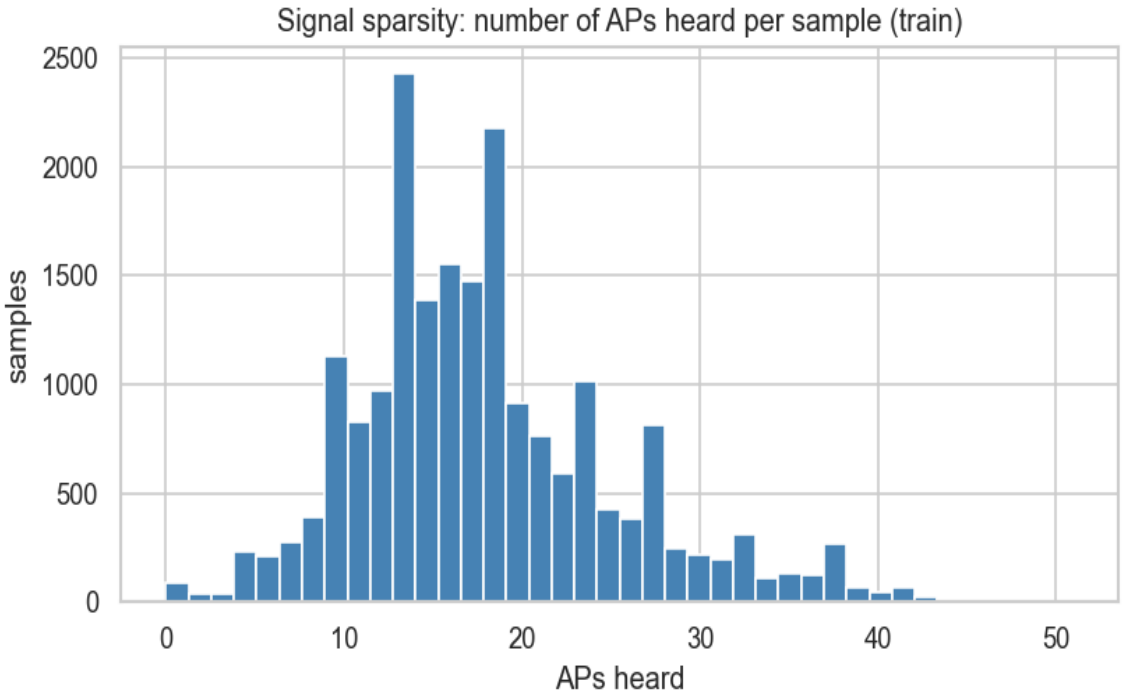
→

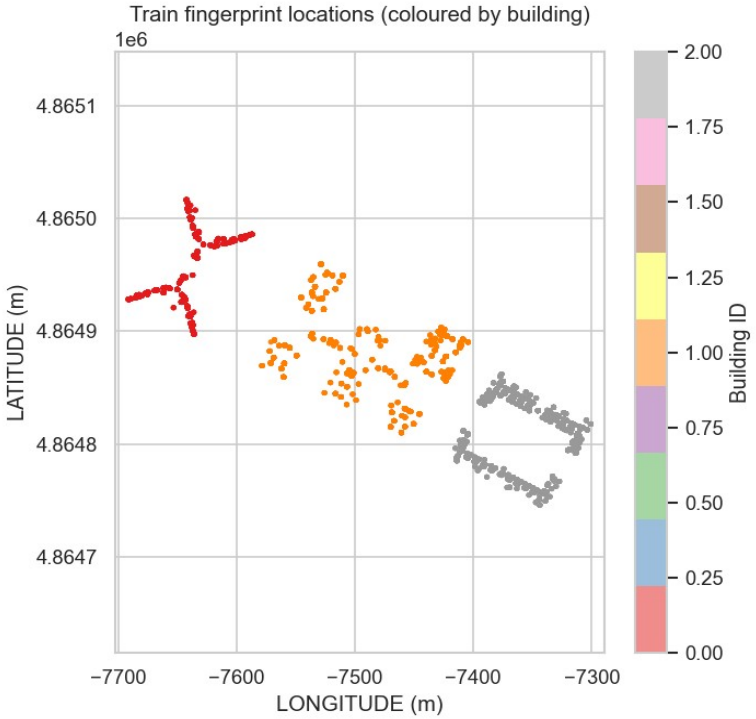
Building	3-class classification
Floor	5-class classification
Position	Two metric coordinates

Benchmark: UJIIndoorLoc (2014). Shared-learning motivation: Caruana (1997).

Source: team code, notebook and verified validation predictions.

Dataset and EDA: sparse signals, uneven floors





96.5% absent · median 17 APs heard

Floor 4 occurs only in Building 2.

19,937 raw train → 19,300 after duplicates | 1,111 validation | supplied split

Source: supplied EDA figures; raw training CSV audit.

Preprocessing and 936-feature representation

465
Scaled RSSI readings

465
Binary absence flags

6
Scan-level features

Feature block	What it represents
5 aggregates	AP count; strongest, mean, standard deviation and range of heard RSSI
1 ordinal feature	Index of strongest kept AP; −1 when no AP is heard

+100 → −105 dBm; remove 55 dead APs; fit scalers on train only. Metadata excluded.

Source: team code, notebook and verified validation predictions.

Bagging averages trees; boosting adds trees sequentially

Random Forest

Many randomised trees
fit in parallel

Average their predictions
to reduce variance.

XGBoost

Add one tree at a time
to improve a regularised loss

Learning rate and tree depth
control each update.

Independent building/floor classifiers + XY regression. RF: 200 trees; XGB: 300, depth 8.

Sources: scikit-learn ensemble guide; XGBoost Introduction to Boosted Trees.

Coordinate error and penalized score are different

Metric	Definition / interpretation
Building / floor accuracy	Correct labels divided by all validation scans
Coordinate MAE	Mean $\lvert\hat{x} - x\rvert$ and mean $\lvert\hat{y} - y\rvert$, separately in metres
Euclidean position error	$e = \sqrt{[(\hat{x} - x)^2 + (\hat{y} - y)^2]}$; report mean, median, P90
Custom combined score	$e + 4 \cdot 1(\text{floor wrong}) + 50 \cdot 1(\text{building wrong})$, in metres

A wrong floor incurs 4 m once, regardless of the number of floors apart.

Source: team code, notebook and verified validation predictions.

Verified results: RF for coordinates, XGB for floors

Model	Building	Floor	Mean e	Median e	P90 e
Random Forest	99.91%	90.73%	11.58 m	6.86 m	24.99 m
XGBoost	99.73%	91.99%	14.52 m	10.06 m	30.38 m
Post-hoc combination	99.91%	91.99%	11.58 m	6.86 m	24.99 m

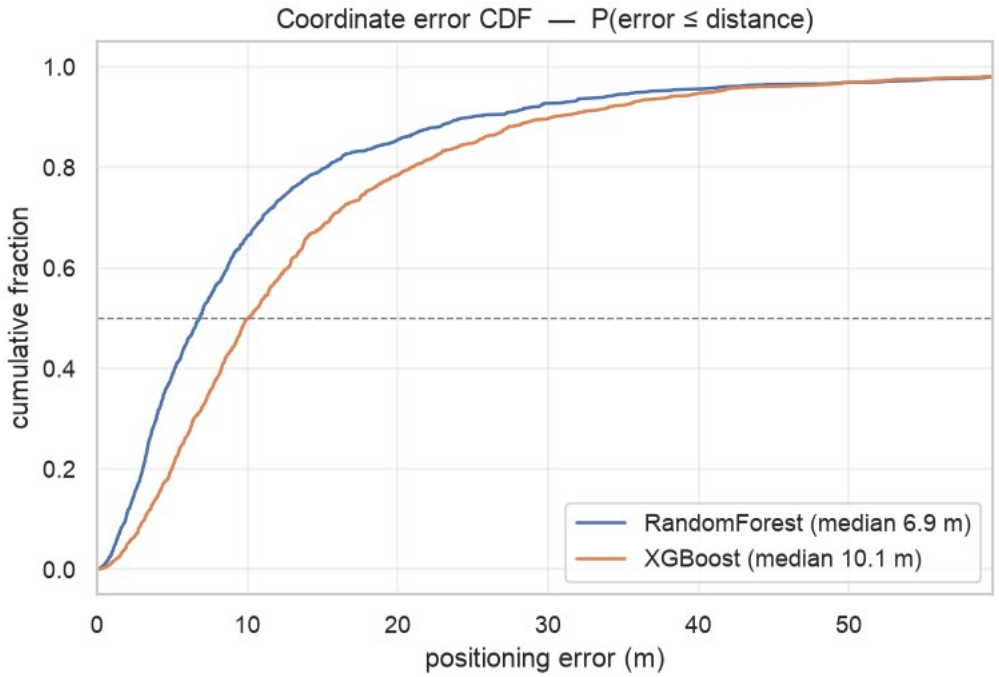
Combination: RF building + XGB floor + RF coordinates.

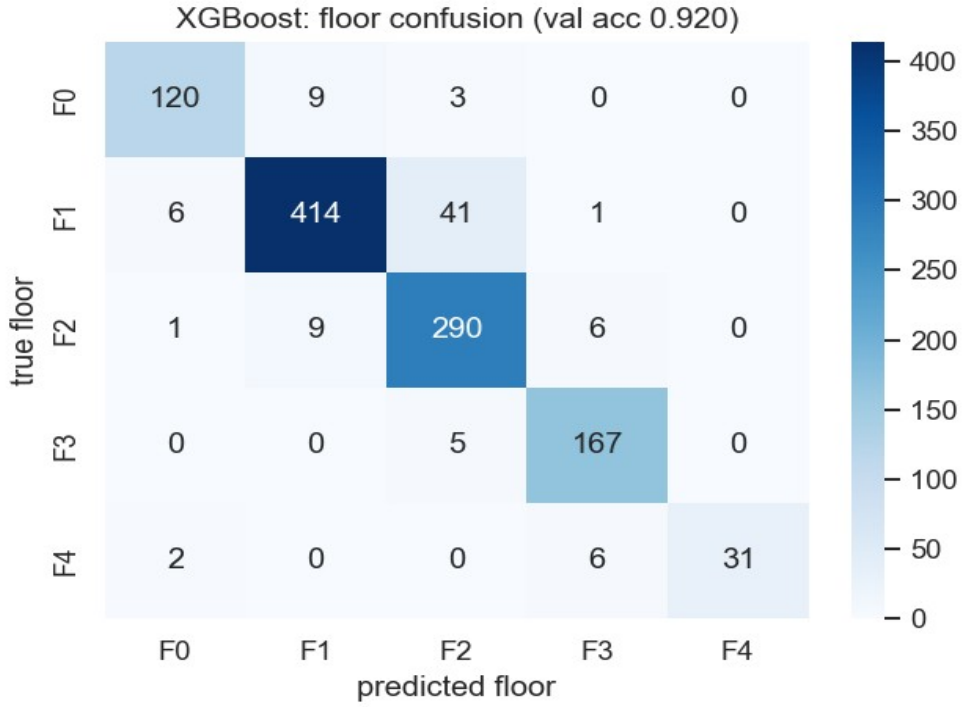
Custom combined score: 11.95 m mean / 7.00 m median.

This is not the Euclidean coordinate error.

Source: team code, notebook and verified validation predictions.

Failure analysis: coordinate tails and rare floors





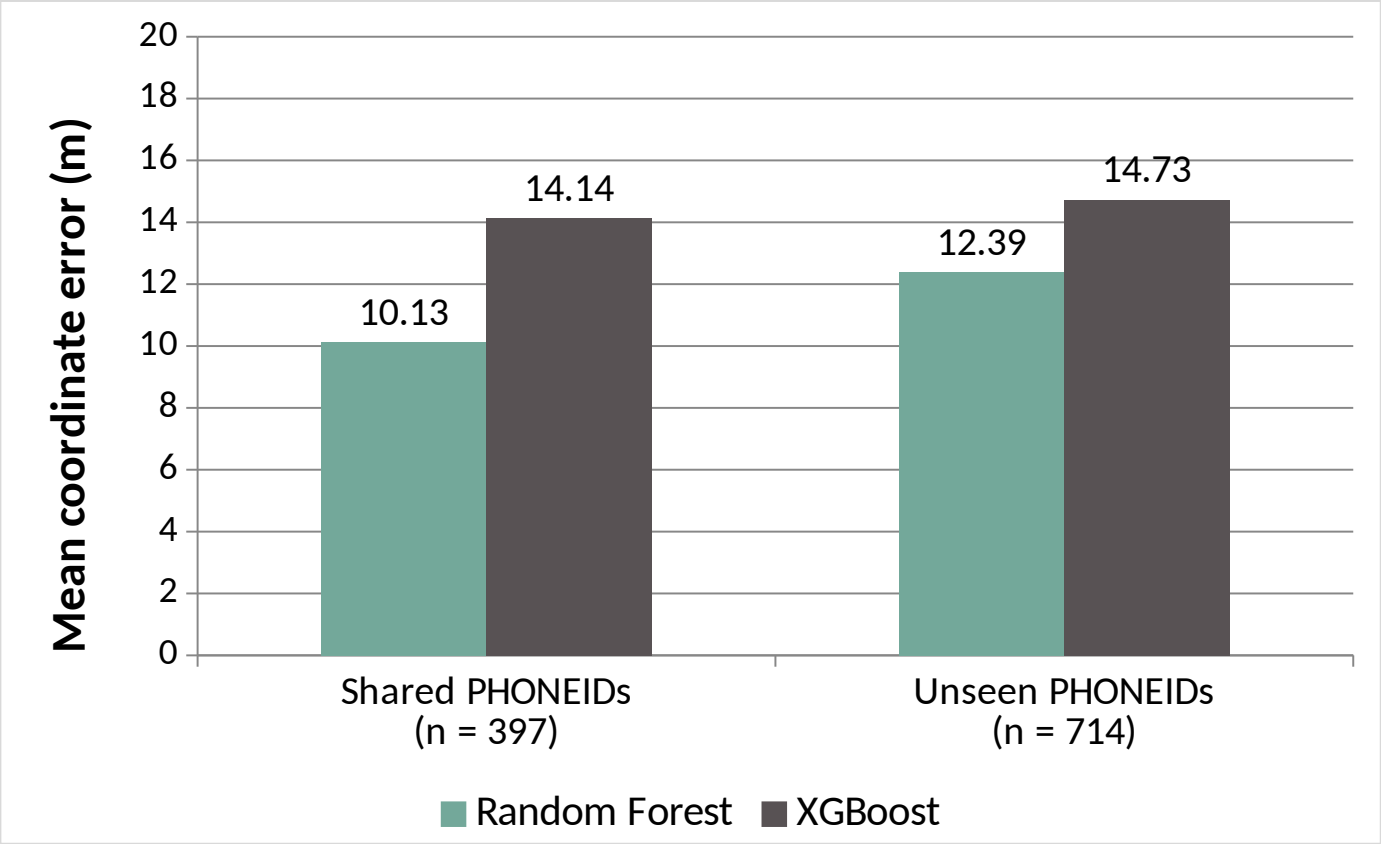
RF: 6.86 m median, 24.99 m P90.

RF error: 10.19 m if floor correct;

25.21 m if wrong. Association, not causation.

Source: team code, notebook and verified validation predictions.

Device analysis: shared versus unseen IDs



IDs 13 and 14 occur in both partitions.

11 validation PHONEIDs

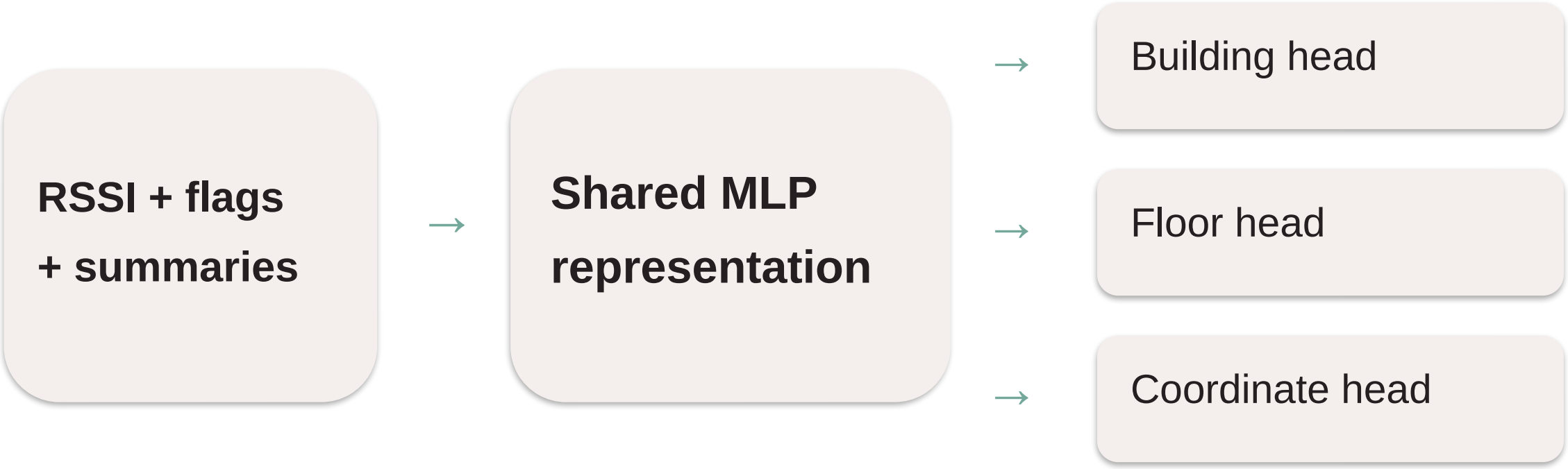
9 absent from training

2 shared with training

Per-phone RF mean errors range from 7.65 m to 30.45 m.

Source: team code, notebook and verified validation predictions.

Limitations and Phase 2: test shared learning



One split, one seed; combination is post-hoc. Phase 2 is proposed, not trained.

Source: official Project 9 brief; Caruana (1997). Proposed work, no DL results.

Takeaway and sources

RF for coordinates; XGB for floors. Device variation and large errors remain.

- Torres-Sospedra et al. (2014). UJIIndoorLoc. IPIN.
  https://doi.org/10.1109/ipin.2014.7275492
- Caruana (1997). Multitask Learning. Machine Learning 28, 41–75.
  https://doi.org/10.1023/a:1007379606734
- UCI dataset: https://archive.ics.uci.edu/dataset/310/ujiindoorloc

- Code: github.com/vedpawar2254/Indoor-location-prediction-using-wifi-signals
  Checked commit b75654b, 27 October 2026.

All result figures refer to the supplied 1,111-row validation set.